Phase 1: Study, Data, and Descriptive Statistics

Sleep duration and dietary supplement use in NHANES, August 2021–August 2023

1. Data Source and Collection Mechanism

This project uses public-use data from the National Health and Nutrition Examination Survey (NHANES), conducted by the National Center for Health Statistics (NCHS), part of the U.S. Centers for Disease Control and Prevention (CDC). The participant data used in this project were collected during the August 2021–August 2023 cycle.

NHANES is a cross-sectional, observational health survey of the U.S. civilian, noninstitutionalized population. It uses a multistage probability-sampling design to select participants from households across the United States. Survey information is collected through interviews and, for relevant components, examinations and laboratory measurements. This project uses the sleep and dietary-supplement questionnaire components; it does not collect new information or assign participants to receive supplements.

Sleep information was collected in the SLQ_L questionnaire by trained interviewers, in the participant's home or by telephone. The supplement-use information in DSQIDS_L covers products reported for the 30 days before the interview. In this NHANES cycle, the supplement interview was conducted by telephone after the first 24-hour dietary recall, and participants were asked to read information from product labels.

The NHANES participant files and their documentation are publicly available through the CDC/NCHS website. The project uses public-use files, so access to the files does not require access to identifiable participant records. However, public files include disclosure protections: some values may be grouped, recoded, or set to missing. In addition, interview participation and self-reported information can limit the completeness and precision of estimates. These data describe reported patterns in one survey cycle; because the study is observational and cross-sectional, they cannot by themselves show that supplement use causes a change in sleep.

Sources: NHANES August 2021–August 2023 sample design and analytic overview; SLQ_L documentation; DSQIDS_L documentation

2. Unit of Observation and Data Structure

The NHANES files used in this project do not all have the same row structure. Some files contain participant-level records, while others contain product or ingredient records.

File	Rows in the downloaded file	What one row represents	Repeated identifiers in this file
DEMO_L	11,933	One NHANES participant, identified by SEQN	None; SEQN is unique
SLQ_L	8,501	One participant's sleep-questionnaire record	None; SEQN is unique
DSQTOT_L	8,860	One participant's total supplement-use record	None; SEQN is unique
DSQIDS_L	11,375	One supplement or antacid product reported by a participant	Yes. There are 4,017 distinct participants; 2,506 have multiple product records, and one participant has up to 30
DSPI	20,235	One supplement product in the product-reference file	None; DSDPID is unique
DSII	188,552	One ingredient entry associated with a supplement product	Yes. A product can have multiple ingredient entries
DSBI	37,061	One component entry associated with a supplement blend	Yes. A blend can have multiple component entries

The row counts above are counts in the downloaded files used for this project. They describe file records, not necessarily the number of participants eligible for the final analysis.

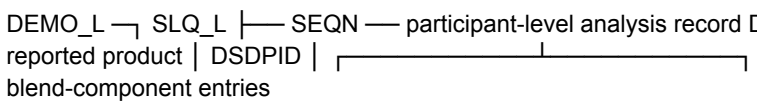
Unit of observation for the main analysis

The participant is the unit of observation in the main analysis. The research question concerns a person's reported sleep duration in relation to that person's supplement-use information. Therefore, the final analysis dataset should contain no more than one row per participant.

The DSQIDS_L file is different: each row represents a reported product, so one participant may appear on multiple rows. Before using these records to describe a participant's supplement exposure, the product reports must be summarized at the participant level. The product and ingredient reference files help identify reported products and their ingredients; they are not additional participant observations.

How the files relate

The participant files are linked using SEQN. Supplement product reports are linked to product and ingredient reference information using DSDPID; blend-component information is linked using DSDIID.



The row counts differ because the files describe different records and may cover different sets of participants or products. A direct merge of all files without checking their structure could create multiple rows for the same participant—for example, when

a participant has several supplement records or a product has several ingredient records. The merge steps must therefore preserve the participant as the analysis unit, and the number of participants and rows should be checked before and after each merge.

Sources: CDC/NCHS documentation for `DEMO_L`, `SLQ_L`, `DSQIDS_L`, `DSQTOT_L`, `DSPI`, `DSII`, and `DSBI`.

3. Sampling, Participant Selection, and Generalizability

NHANES does not use a simple random sample of all U.S. residents. It uses a complex, multistage probability-sampling design to select people from the U.S. civilian, noninstitutionalized population. In general, geographic areas are selected first, followed by smaller area segments, dwelling units, and then eligible people within households. This design gives people different probabilities of selection and is intended to support estimates for the target population when the appropriate survey weights and design information are used.

The August 2021–August 2023 cycle used an updated sample design following the interruption of NHANES field operations during the COVID-19 pandemic. In this cycle, NHANES did not oversample participants by race and Hispanic origin or income. Selection within households also varied by age: all encountered household members aged 0–19 years and 60 years or older were selected, while the number selected among adults aged 20–59 depended on the household size. These design features mean that the sample composition and precision for some subgroups may differ from those in earlier NHANES cycles.

NHANES is observational: participants were not randomly assigned to use dietary supplements. For this project, the planned analysis focuses on adults aged 20 years or older who have the information needed to examine reported supplement use and sleep duration. Although the sleep questionnaire was available to participants aged 16 years or older, the project's adult-age restriction narrows the population represented by the analysis.

The NHANES design is intended to support generalization to the U.S. civilian, noninstitutionalized population, but results from this project can be generalized only with appropriate consideration of the survey design, weights, participation, and missing data. The final analytic sample will also be limited to people who meet the project's eligibility rules and have the required sleep and supplement information. Therefore, the final conclusions will apply most directly to eligible adults represented by the analyzed data, rather than automatically to every U.S. resident or to people outside the NHANES target population. The exact number of people retained at each eligibility and data-availability step will be reported when the analytic population is finalized.

Because supplement use was not assigned experimentally and the data are cross-sectional, any observed difference or association between supplement use and sleep should be interpreted as an association, not as evidence that supplement use caused a change in sleep.

Sources: NHANES 2021–2023 sample design and analytic overview; NHANES demographics and sample weights documentation; NHANES sleep questionnaire documentation.

4. Classification of Main Variables and Measurement Scales

The main outcome variables are usual sleep duration on weekdays and weekends. The main explanatory variable is dietary supplement use. At this preliminary stage, supplement use is defined broadly as reported use versus no reported use during the reference period. If the analysis is later narrowed to a specific supplement type or ingredient, that exposure will be defined at the participant level using the reported-product file and the NHANES product and ingredient reference files.

Variable	NHANES variable or source	Role in the project	Variable type and measurement scale
Usual weekday sleep duration	`SLD012` in `SLQ_L`	Outcome	Quantitative numerical variable, recorded in hours in half-hour increments for most values; ratio scale
Usual weekend sleep duration	`SLD013` in `SLQ_L`	Outcome	Quantitative numerical variable, recorded in hours in half-hour increments for most values; ratio scale
Any dietary supplement use	Participant-level use information in `DSQTOT_L`	Main explanatory or grouping variable	Categorical, binary, nominal: reported use / no reported use
Supplement type or ingredient, if selected for a narrower analysis	Derived from `DSQIDS_L` and the `DSPI`, `DSII`, and `DSBI` reference files	More specific explanatory or grouping variable	Categorical, nominal; a participant may belong to more than one ingredient group if they reported multiple products
Age at screening	`RIDAGEYR` in `DEMO_L`	Descriptive or potential explanatory variable; also used to define adult eligibility	Quantitative, recorded in whole years; ratio scale
Sex category as recorded by NHANES	`RIAGENDR` in `DEMO_L`	Descriptive or potential grouping variable	Categorical, binary, nominal
Dietary Day 1 sample weight	`WTDRD1` in `DSQTOT_L`	Survey-weight variable for consideration in population estimates	Quantitative numerical weight; a sampling-design variable, not a health outcome
Survey stratum and primary sampling unit	`SDMVSTRA` and `SDMVPSU` in `DEMO_L`	Survey-design variables for variance estimation	Categorical design identifiers; they are not substantive health variables
Participant identifier	`SEQN`	Used to link participant files and check uniqueness	Identifier only; nominal label, not a quantitative analysis variable

For the preliminary research question, sleep duration is the response (outcome) variable, and supplement-use status is the explanatory or grouping variable. Age and sex may be used to describe the analytic sample or to form descriptive subgroups. The project will not treat the participant identifier or survey-design identifiers as measured health characteristics.

The sleep-duration variables require special care. NHANES reports values of 2 as “less than 3 hours” and values of 14 as “14 hours or more.” Therefore, these endpoint values are grouped categories rather than exact sleep durations. They must not automatically be interpreted as exactly 2 or 14 hours in summaries that assume exact numeric values. Age is also top-coded at 80 years in the public-use file, so a recorded age of 80 may mean 80 years or older.

The final definition of the supplement exposure and the exact sample weight will be confirmed before the main analysis. The appropriate weight and survey-design variables will be justified in the NHANES-specific sampling and weighting section.

5. Data Quality Audit

An initial data-quality audit was conducted on the downloaded NHANES files, focusing on participant and reference identifiers, repeated records, missing values, documented special codes, plausible ranges, and key file links. The original XPT files were retained unchanged. Any recoding or exclusions for analysis will be documented in a separate analysis dataset.

Check	Findings in the downloaded files	Decision for analysis
Participant identifiers	`SEQN` is present and unique in `DEMO_L` (11,933 records), `SLQ_L` (8,501 records), and `DSQTOT_L` (8,860 records). No missing `SEQN` values were found in these files.	Use `SEQN` to link participant-level files and verify that each participant has no more than one row after person-level merging.
Repeated supplement records	`DSQIDS_L` contains 11,375 product records from 4,017 participants. A participant can report multiple products: 2,506 participants have more than one record, and the maximum is 30 records for one participant. One participant-product pair uses the same unknown-product identifier in two records with different reported intake details.	Multiple product records per person are expected and must not be treated as repeated sleep measurements. The single repeated unknown-product identifier will be flagged and not automatically deleted.
Product and ingredient reference links	`DSPI` contains 20,235 unique `DSDPID` product identifiers. All 188,552 `DSII` ingredient records link to a product in `DSPI`; all 37,061 `DSBI` blend-component records link to an ingredient identifier in `DSII`.	Keep reference records separate from participant counts. Check identifier matches after each link so ingredient rows do not multiply participant rows in the analysis dataset.
Product-identification status	In `DSQIDS_L`, 11,210 of 11,375 report records have product matches with codes 1–5, which range from exact or near-exact to default matches. The remaining 165 records have code 6 (“no match,” 158 records) or code 9 (“don’t know,” 7 records); these involve 147 distinct product identifiers.	Do not assign an ingredient to an unmatched or unknown product. Retain the reported use information where relevant, but classify the product or ingredient exposure as unresolved for ingredient-specific analyses.
Sleep values and missingness	`SLD012` has 113 missing values and `SLD013` has 114 missing values among 8,501 records. NHANES codes a value of 2 as “less than 3 hours” and 14 as “14 hours or more”; these are grouped endpoints, not exact durations.	Keep missing values distinct from reported sleep durations. Do not interpret endpoint codes 2 and 14 as exact hours without an explicitly documented method.
Supplement-use status and survey weight	In `DSQTOT_L`, `DSD010` uses 1 = yes, 2 = no, 7 = refused, and 9 = don’t know. There are also missing values. `WTDRD1 = 0` indicates that the Day 1 dietary recall was not done or was incomplete; it is not an ordinary positive survey weight.	Do not recode refused, don’t-know, or missing status as “no.” Apply the documented weight rules when defining the analytic sample and estimating population quantities.

Participant identifiers	SEQN is present and unique in DEMO_L (11,933 records), SLQ_L (8,501 records), and DSQTOT_L (8,860 records). No missing SEQN values were found in these files.	Use SEQN to link participant-level files and verify that each participant has no more than one row after person-level merging.
Supplement detail special codes	In DSQIDS_L, DSD103 has 6 missing values and 28 “don’t know” codes (9999). DSD122Q has 5 missing values and 2 “don’t know” codes (999999). The observed ordinary values fall within the ranges documented by NHANES.	Treat special codes as non-substantive responses, not as actual days or quantities. Preserve the raw values and recode only in the analysis copy.
Age and category codes	RIDAGEYR has no missing values in DEMO_L, ranges from 0 to 80, and NHANES top-codes ages 80 and older as 80. RIAGENDR contains the documented categories 1 and 2, with no missing values in this file.	Treat age 80 as “80 years or older,” not necessarily exactly 80. Treat RIAGENDR as a categorical variable, not a numeric measurement.

Overall, the identifier and reference-file checks were consistent with the documented file structure. Repeated participant records in DSQIDS_L and repeated product or ingredient identifiers in the reference files are expected because these files contain product-level and ingredient-level records. They are not, by themselves, evidence of data-entry errors. The main risks for analysis are mishandling special codes, treating missing or unresolved information as a negative response, and allowing multi-row product or ingredient links to create duplicate participant rows.

Before producing final results, the analysis code will report how each special code and missing value is handled, verify the participant count and uniqueness after every merge, and document the number of participants excluded at each eligibility step.

Sources: NHANES DEMO_L documentation, NHANES SLQ_L documentation, NHANES DSQIDS_L documentation, and NHANES DSQTOT_L documentation.

6. Frequency Tables and Empirical Distributions

The following are preliminary, unweighted descriptive summaries for the 6,064 participants aged 20 years or older in the participant-level dataset created by linking DEMO_L, SLQ_L, and DSQTOT_L using SEQN. Each participant contributes at most one row to this dataset. The number of valid observations differs by variable because some responses are missing or were recorded as refused or "don't know." These summaries describe the current analysis dataset and are not population estimates.

Categorical variables

Variable	Category	Frequency (n)	Percent
Age group	20–39 years	1,559	25.7%
Age group	40–59 years	1,733	28.6%
Age group	60 years or older	2,772	45.7%
NHANES sex category	Category 1	2,720	44.9%
NHANES sex category	Category 2	3,344	55.1%
Supplement use (DSD010)	Yes	3,092	64.7% of valid yes/no responses
Supplement use (DSD010)	No	1,687	35.3% of valid yes/no responses

The age-group and sex percentages use all 6,064 adults as the denominator. For supplement use, the percentages are calculated among the 4,779 participants with a valid "yes" or "no" response. In the full adult dataset, DSD010 also has 10 refused responses, 4 "don't know" responses, and 1,271 missing values. These responses are not classified as "no."

Empirical distributions of sleep duration

NHANES records the sleep-duration variables in hours, usually in half-hour increments. To summarize the observed distributions, values were grouped as less than 7 hours, 7–8.5 hours, or 9 hours or more. The percentages below use the number of valid responses for each sleep variable.

Sleep duration	Less than 7 hours	7–8.5 hours	9 hours or more	Valid n
Weekdays/workdays (SLD012)	1,269 (21.2%)	3,350 (55.8%)	1,381 (23.0%)	6,000
Weekends/non-workdays (SLD013)	816 (13.6%)	2,860 (47.7%)	2,322 (38.7%)	5,998

The largest group reported 7–8.5 hours of sleep on weekdays. Weekend reports were more spread toward longer durations: a larger share fell in the 9-hours-or-more group, and a smaller share fell below 7 hours, compared with weekday reports. This is a descriptive pattern only; it does not establish that people sleep significantly longer on weekends or explain why the pattern occurs.

NHANES codes 2 as "less than 3 hours" and 14 as "14 hours or more." These endpoint responses were included in the broad groups above as less than 7 hours and 9 hours or more, respectively; they were not treated as exact values of 2 or 14 hours. Missing sleep responses were excluded from the denominator for each sleep variable.

Sources: NHANES documentation for SLQ_L and DSQTOT_L. Counts and percentages were calculated from the project's downloaded XPT files.

7. Measures of Center and Spread

The summaries below use the same preliminary, unweighted dataset as in Section 6: participants aged 20 years or older with records in the linked `DEMO_L`, `SLQ_L`, and `DSQTOT_L` files. The number of valid observations varies by variable. These are descriptive summaries of the current dataset, not weighted population estimates.

Variable	Valid n	Mean*	SD*	Median	Q1	Q3	IQR	Observed endpoints in the released data
Weekday/workday sleep (`SLD012`), hours	6,000	7.70	1.59	8.0	7.0	8.5	1.5	2 = less than 3 hours; 14 = 14 hours or more
Weekend/non-workday sleep (`SLD013`), hours	5,998	8.26	1.71	8.0	7.5	9.0	1.5	2 = less than 3 hours; 14 = 14 hours or more
Age at screening (`RIDAGEYR`), years	6,064	53.85	17.18	57.0	39.0	68.0	29.0	20 to 80; 80 represents 80 years or older

For the sleep variables, the mean and standard deviation were calculated using the numeric values as released by NHANES. Because the endpoint codes 2 and 14 represent grouped durations rather than exact values, these mean and SD estimates are approximate summaries of the released codes. The median and IQR are emphasized because they describe the center and middle spread without relying strongly on those extreme endpoint values.

The weekday and weekend sleep medians are both 8 hours, and both IQRs are 1.5 hours. The weekend mean based on the released codes is higher than the weekday mean, and the weekend upper quartile is 9 hours compared with 8.5 hours on weekdays. These are descriptive differences only; no inferential claim is made at this stage.

For age, the mean is slightly lower than the median, and the middle half of the sample falls between 39 and 68 years. Because NHANES top-codes ages of 80 years or older as 80, the reported maximum should be interpreted as “80 or older,” not necessarily exactly 80.

Sources: NHANES sleep questionnaire documentation and NHANES demographics documentation. Statistics were calculated from the project’s downloaded XPT files.

8. Appropriate Visualization

Four unweighted figures were prepared for adults aged 20 years or older with a valid yes/no response for supplement use. The figures do not require a positive `WTDRD1` weight. The first two show the empirical distributions of usual weekday and weekend sleep duration. The other two compare sleep duration by reported supplement-use status. This exploratory figure sample is therefore slightly larger than the positive-weight analytic sample defined in A4.

The weekday and weekend distribution plots use percentages and show each reported sleep-duration value. NHANES endpoint codes are labelled as “<3” and “14+” because they represent grouped durations, not exact values of 2 and 14 hours. The supplement-group boxplots compare only reported numeric sleep values from 3 to 13.5 hours; the grouped endpoint categories are excluded. Outlier markers are omitted from the boxplots for readability.

The weekday boxplots show the same median and quartiles in both supplement-use groups. For weekend sleep, the medians are also the same, while the lower quartile is slightly higher among participants reporting supplement use. The distributions overlap substantially. These figures are descriptive and unweighted; they do not demonstrate a statistically significant difference, a causal effect, or a population-level association.

Figures:

• Weekday/workday sleep-duration distribution among adults aged 20 years or older.

• Weekend/non-workday sleep-duration distribution among adults aged 20 years or older.

• Weekday/workday sleep duration by reported supplement-use status.

• Weekend/non-workday sleep duration by reported supplement-use status.

Source: NHANES sleep questionnaire documentation for the interpretation of sleep-duration values (SLQ_L documentation).

Figures

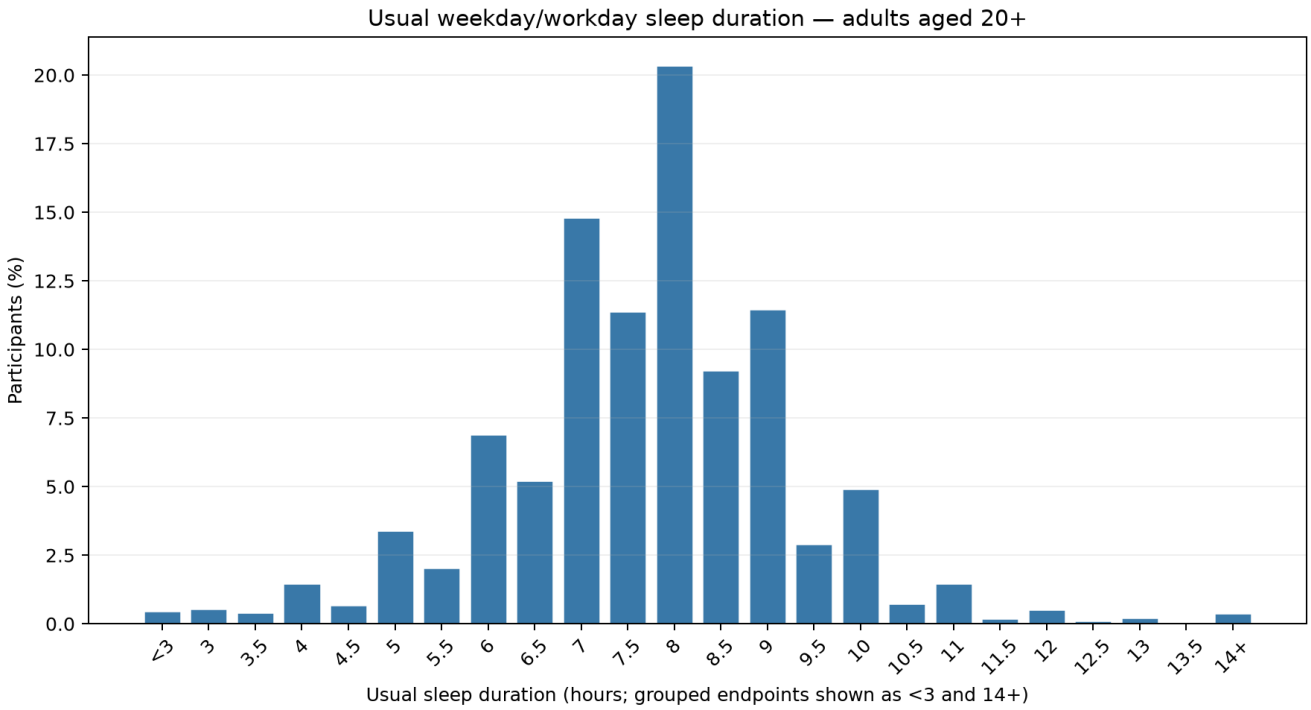


Figure 1. Usual weekday/workday sleep-duration distribution among adults aged 20 years or older.

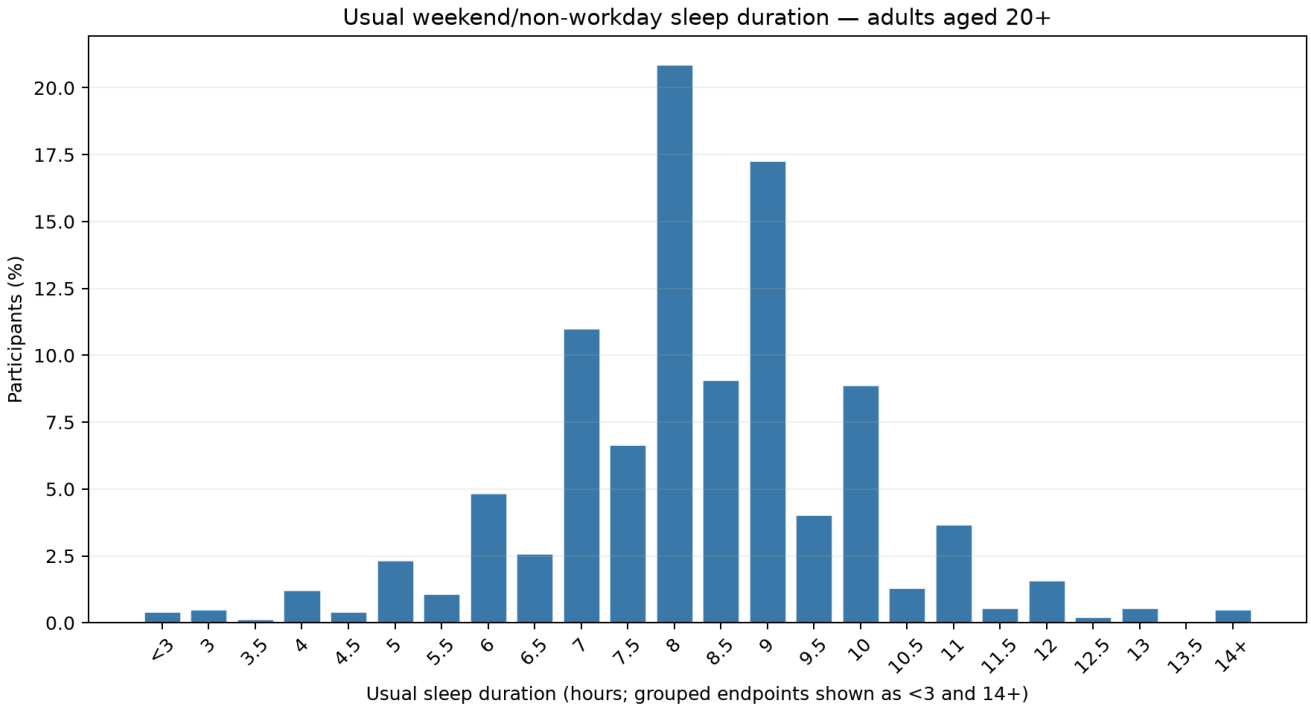


Figure 2. Usual weekend/non-workday sleep-duration distribution among adults aged 20 years or older.

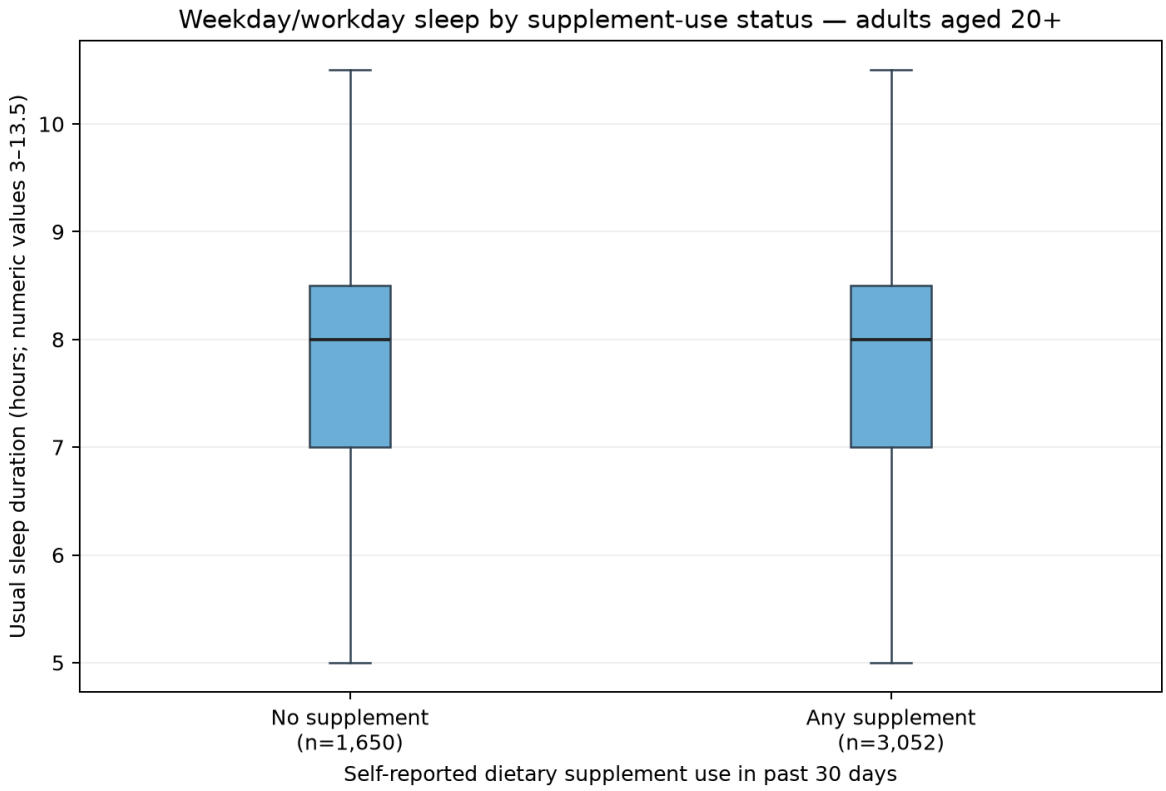


Figure 3. Weekday/workday sleep duration by reported supplement-use status.

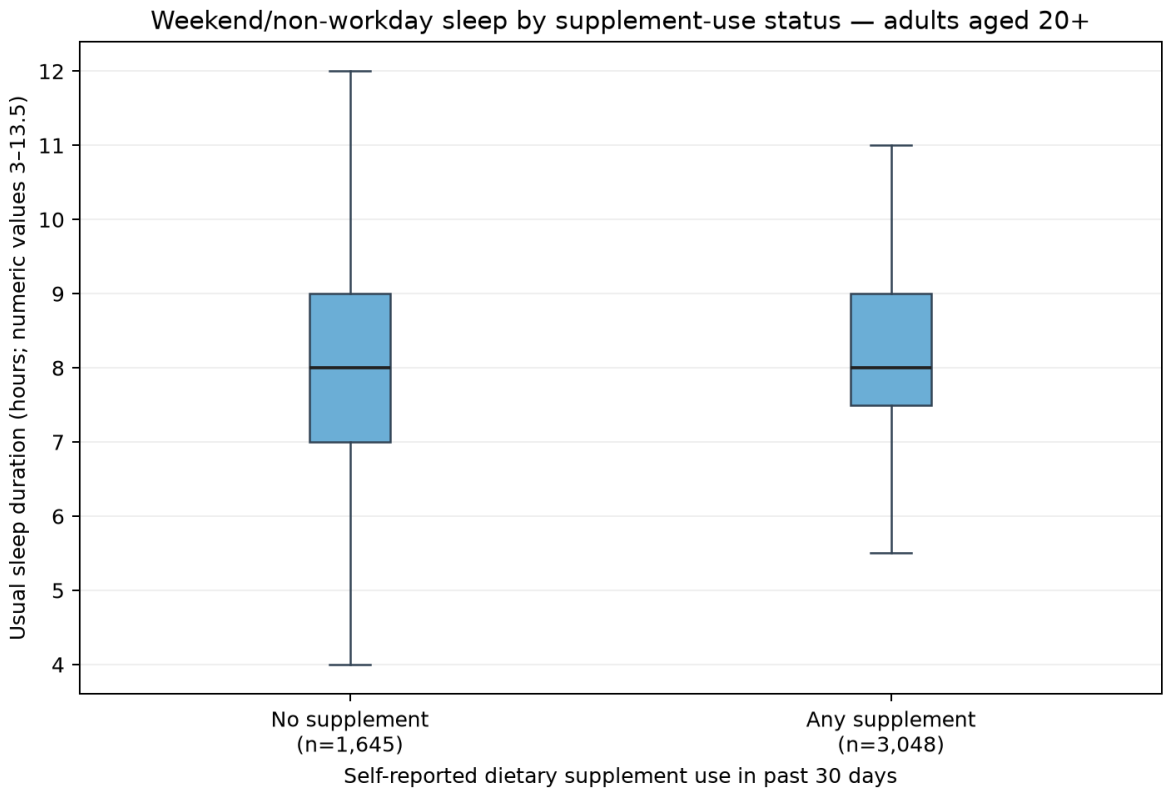


Figure 4. Weekend/non-workday sleep duration by reported supplement-use status.

9. Preliminary Descriptive Comparison

Usual sleep duration was compared between adults who reported dietary supplement use in the past 30 days and adults who reported no use. The comparison is unweighted and descriptive. It includes adults with a valid yes/no response for `DSD010`, regardless of whether `WTDRD1` is positive. Thus, the weekday comparison includes 4,702 participants and the weekend comparison includes 4,693. A4 reports the slightly smaller positive-weight analytic samples (4,699 and 4,690, respectively), which are used for the weighted summaries in A3. It uses reported sleep values from 3 to 13.5 hours; missing sleep values and the NHANES grouped endpoint categories (“less than 3 hours” and “14 hours or more”) are excluded because the endpoint codes do not represent exact durations.

Sleep variable	Supplement use reported: n	Median (Q1, Q3)	No supplement use reported: n	Median (Q1, Q3)
Weekday/workday sleep, hours	3,052	8.0 (7.0, 8.5)	1,650	8.0 (7.0, 8.5)
Weekend/non-workday sleep, hours	3,048	8.0 (7.5, 9.0)	1,645	8.0 (7.0, 9.0)

The weekday sleep medians and quartiles are the same in both groups. For weekend sleep, the median is also 8 hours in both groups; the lower quartile is slightly higher among participants reporting supplement use, while the upper quartile is the same. Overall, the descriptive summaries and boxplots show substantial overlap between the groups and no clear separation in their central sleep durations.

This comparison does not establish that the groups differ in the population, and it does not show that supplement use causes a change in sleep. The exposure is also broad: it indicates any reported supplement use, rather than use of a specific supplement or ingredient. The comparison should therefore be interpreted as an initial description of the current sample, not as a conclusion about a particular supplement’s relationship with sleep.

Source for the grouped sleep-duration codes: NHANES `SLQ_L` documentation.

10. Finalizing the Statistical Question

Final statistical question

Among adults aged 20 years or older represented in NHANES August 2021–August 2023, how do the distributions of usual weekday/workday sleep duration and usual weekend/non-workday sleep duration compare between participants who reported using any dietary supplement in the past 30 days and participants who reported no dietary supplement use?

Why this question fits the available data

The question uses participant-level information that can be linked by the NHANES respondent identifier, `SEQN`. The supplement-use file, `DSQTOT_L`, identifies participants who reported any supplement use and those who reported none. The sleep questionnaire file, `SLQ_L`, provides usual sleep-duration measures for weekdays/workdays (`SLD012`) and weekends/non-workdays (`SLD013`). The demographic file, `DEMO_L`, provides age for defining the adult population.

The preliminary merge contained 6,064 adults aged 20 years or older. Among them, 4,779 had a valid yes/no response for the any-supplement-use measure. The number included in each sleep comparison will depend on whether the participant also has a valid value for that sleep outcome. The final analysis counts and the handling of NHANES sampling weights and design variables will be documented in the relevant Path A sections.

Statistical nature and interpretation

This is a descriptive comparison and an observational association question. The participant is the unit of observation. The analysis will compare the observed sleep-duration distributions of supplement users and non-users; it will not estimate a causal effect of supplements on sleep.

In the preliminary unweighted comparison, the median sleep duration was 8 hours in both groups for both weekday/workday and weekend/non-workday sleep. This indicates that the medians alone do not show a clear difference. It does not establish that the groups are identical: the full distributions and properly weighted estimates should be examined before drawing conclusions.

Scope and limitations

“Any dietary supplement use” combines many different products and ingredients, so the exposure is broad and does not identify which supplement, dose, or reason for use is associated with sleep duration. NHANES 2021–2023 did not collect respondents’ reasons for taking supplements; therefore, this project cannot determine whether participants used a supplement to improve sleep. In addition, the data are observational and cross-sectional, so the results will be interpreted as differences or associations, not as evidence that supplement use causes a change in sleep.

A melatonin-specific comparison may be considered as an exploratory secondary analysis. The preliminary data identified about 101 adult melatonin reporters with a valid yes/no supplement-use response, so that subgroup may be too small for precise population estimates. Any such analysis will be treated cautiously and its precision assessed using the appropriate NHANES design.

Preliminary expectation

Before conducting the final analysis, no directional difference is assumed. The preliminary unweighted medians were the same in the two broad supplement-use groups, so the analysis will examine whether other features of the distributions differ and whether the findings change after applying the appropriate NHANES weights.

A1. Files and Merge Key

The participant-level analysis dataset was assembled from the NHANES August 2021–August 2023 files DEMO_L, SLQ_L, and DSQTOT_L. The common participant identifier is SEQN.

Each of these three files contains one record per participant and has no repeated SEQN values. The files were linked sequentially using SEQN. Starting with DEMO_L, 11,933 participants were present. Of these, 8,501 had a matching record in SLQ_L. Linking those participants to DSQTOT_L retained 6,615 participants with records in all three files.

Merge stage	Participants before merge	Participants matched and retained	Participants without a matching record
DEMO_L starting file	11,933	11,933	—
Add SLQ_L using SEQN	11,933	8,501	3,432
Add DSQTOT_L using SEQN	8,501	6,615	1,886

The resulting three-file dataset contains 6,615 unique participants. Applying the adult age criterion of 20 years or older using RIDAGEYR in DEMO_L retains 6,064 participants in this merged dataset. The final number for each analysis may be lower when a participant is missing a value for a required exposure or sleep outcome.

DSQIDS_L contains individual supplement and antacid reports. It has 11,375 rows for 4,017 distinct participants, and some participants have multiple product records. In the downloaded file, one participant has as many as 30 records. Therefore, raw DSQIDS_L rows should not be directly joined to the participant-level analysis table, because that could count some participants multiple times. The product records can be summarized into participant-level exposure variables before being joined using SEQN.

The product-reference files are not participant-level files. DSQIDS_L links to product information in DSPI through DSDPID. Ingredient records in DSII are also associated with products through DSDPID; blend information can be linked from DSII to DSBI through DSDIID. These reference-file links can produce multiple rows per product or ingredient, so they must be handled separately from the one-row-per-participant merge.

Participant counts above were calculated from the downloaded XPT files. The counts describe records available in those files; a missing match in one component file does not, by itself, establish why that participant has no record there.

Sources

- NHANES demographics and sample weights (DEMO_L)
- NHANES sleep questionnaire (SLQ_L)
- NHANES total dietary supplements (DSQTOT_L)
- NHANES individual dietary supplements (DSQIDS_L)
- NHANES supplement product information (DSPI)
- NHANES supplement ingredient information (DSII)
- NHANES supplement blend information (DSBI)

A2. NHANES Sample Weights and Design Variables

Selected sample weight

The primary sample weight for this project is WTDRD1, the dietary day one sample weight. The main exposure is 30-day dietary supplement use from DSQTOT_L. NHANES documentation specifically instructs analysts to use WTDRD1 when analyzing the 30-day dietary supplement data.

In the August 2021–August 2023 cycle, the supplement-use interview was conducted after the first 24-hour dietary recall. WTDRD1 starts from the Mobile Examination Center sample weight and includes further adjustments for additional nonresponse and for the different allocation of dietary interviews across weekdays and weekends. It is therefore the appropriate weight for the supplement-use component of this analysis.

The sleep measures are from the SLQ_L questionnaire, which does not provide a separate sleep-specific sample weight. Because the analysis combines sleep information with the 30-day supplement-use component, the analysis will use the weight specified by NHANES for the supplement data, WTDRD1. The analytic dataset must be restricted to participants with a positive WTDRD1 and valid values for the variables included in each analysis.

Survey design variables

The design variables are:

• SDMVSTRA: masked variance pseudo-stratum

• SDMVPSU: masked variance pseudo-primary sampling unit

These variables are provided in DEMO_L and are linked to the participant-level analysis dataset using SEQN. The sample weight is used when calculating population estimates. The strata and PSU variables are used with the weight to account for the NHANES sampling design when estimating uncertainty, such as standard errors and confidence intervals.

Why weighting is needed

NHANES uses a stratified, clustered, multistage probability sample rather than a simple random sample. Participants can have different probabilities of selection and different response patterns. A sample weight represents the number of people in the target population represented by a sampled participant, after the relevant selection and nonresponse adjustments. Accordingly, weighted estimates are needed for population-level interpretation, while the strata and PSU variables are needed to account for the design when estimating uncertainty.

Preliminary weight-availability check

The downloaded DSQTOT_L file contains 6,754 positive WTDRD1 values. The codebook identifies a value of zero for participants whose Day 1 dietary recall was not done or was incomplete. In the merged adult dataset of 6,064 participants, 4,798 have a positive WTDRD1; 4,775 of those also have a valid yes/no response for DSD010. These are preliminary availability counts, not the final sample size for either sleep analysis. The final number will depend on the sleep outcome being analyzed and the project's eligibility and missing-data rules.

NHANES advises analysts to use the appropriate component weight and design information and notes that weight choice should reflect the most restrictive data component used in a combined analysis. WTDRD1 is selected here because the main supplement-use exposure comes from the dietary supplement component and the cycle-specific DSQTOT_L documentation explicitly identifies this weight for 30-day supplement analyses.

Sources

- NHANES DSQTOT_L documentation and analytic notes
- NHANES August 2021–August 2023 sample design and analytic guidelines
- NHANES DEMO_L documentation
- NHANES SLQ_L documentation

A3. Comparison of Unweighted and Weighted Summaries

To illustrate the effect of weighting, the unweighted and weighted prevalence of any dietary supplement use were compared among adults aged 20 years or older with a positive `WTDRD1` weight and a valid yes/no response to `DSD010`.

Summary	Unweighted estimate	`WTDRD1`-weighted estimate
Adults reporting any dietary supplement use (`DSD010 = 1`)	64.7% (3,090 of 4,775)	60.2%

The weighted estimate is 4.5 percentage points lower than the unweighted estimate. This shows that the sample proportion of supplement users differs from the population estimate after accounting for the NHANES dietary sample weights. The weights account for unequal selection probabilities, additional nonresponse, and the weekday/weekend allocation of dietary interviews.

As a project-related comparison, mean sleep duration was also summarized by supplement-use group. Only sleep-duration values from 3 to 13.5 hours were used for these means because NHANES codes the lower and upper values as grouped categories (`2` means less than 3 hours; `14` means 14 hours or more), rather than exact sleep durations.

Sleep measure and group	Unweighted mean	`WTDRD1`-weighted mean	Valid participants
Weekday/workday sleep — supplement use reported	7.76 hours	7.71 hours	3,051
Weekday/workday sleep — no supplement use reported	7.61 hours	7.60 hours	1,648
Weekend/non-workday sleep — supplement use reported	8.23 hours	8.29 hours	3,047
Weekend/non-workday sleep — no supplement use reported	8.23 hours	8.29 hours	1,643

For weekday/workday sleep, the unweighted difference between the two groups was about 0.15 hours, compared with about 0.11 hours after weighting. For weekend/non-workday sleep, the weighted group means were nearly identical. These results are descriptive point estimates. They do not establish that the groups differ statistically, and they do not imply a causal effect of supplement use on sleep.

The unweighted summaries describe the participants in the analyzed sample. The weighted summaries account for the NHANES dietary sample weights and are more appropriate for population-level interpretation. This comparison is an introductory demonstration of the effect of weighting; it does not report design-based standard errors or confidence intervals.

The weighted supplement-use proportion was calculated as the sum of `WTDRD1` among participants reporting supplement use divided by the sum of `WTDRD1` among participants with a valid yes/no response. Weighted sleep means were calculated within each supplement-use group as the weighted sum of sleep hours divided by the sum of `WTDRD1` for participants with valid sleep hours in that group.

Sources: NHANES recommends `WTDRD1` for analyses of 30-day dietary supplement use and explains that it accounts for additional nonresponse and weekday/weekend allocation of dietary interviews (`DSQTOT_L` analytic notes). NHANES also describes the cycle's sample design and guidance on selecting weights for analyses that combine components (2021–2023 analytic guidelines).

A4. Definition of the Analytic Population

Eligibility criteria

The primary analytic population consists of NHANES participants who meet all of the following criteria:

• Have records in DEMO_L, SLQ_L, and DSQTOT_L, linked by SEQN.

• Are aged 20 years or older, based on RIDAGEYR in DEMO_L.

• Have a positive WTDRD1 dietary sample weight.

• Have a valid yes/no response to DSD010 in DSQTOT_L.

• Have a usable value for the sleep outcome being analyzed.

For the sleep-duration means, only values from 3 to 13.5 hours are treated as exact numeric values. NHANES uses 2 to represent less than 3 hours and 14 to represent 14 hours or more. These grouped categories, and missing sleep values, are excluded from analyses of the numeric mean.

Participant flow

Selection step	Participants retained	Excluded at this step
DEMO_L starting file	11,933	—
Matched to SLQ_L using SEQN	8,501	3,432
Matched to DSQTOT_L using SEQN	6,615	1,886
Aged 20 years or older	6,064	551
Positive WTDRD1 weight	4,798	1,266
Valid yes/no response to DSD010	4,775	23

The number of participants with a valid sleep-duration value depends on the outcome:

Sleep outcome	Participants with usable sleep data	Supplement use reported (DSD010 = 1)	No supplement use reported (DSD010 = 2)	Excluded for missing or grouped sleep value
Weekday/workday sleep (SLD012)	4,699	3,051	1,648	76
Weekend/non-workday sleep (SLD013)	4,690	3,047	1,643	85

Among the 76 participants excluded from the weekday/workday numeric-mean analysis, 38 had a missing sleep value, 23 had the grouped value for less than 3 hours, and 15 had the grouped value for 14 hours or more. Among the 85 participants excluded from the weekend/non-workday numeric-mean analysis, 42 had a missing value, 20 had the grouped value for less than 3 hours, and 23 had the grouped value for 14 hours or more.

A total of 4,666 participants have usable numeric values for both sleep outcomes. The weekday and weekend outcomes will otherwise be analyzed using their outcome-specific sample sizes shown above.

Handling of exclusions

Participants younger than 20 years are excluded because the project's target population is adults. Participants with a zero WTDRD1 are excluded from the weighted analytic population; NHANES identifies a zero dietary day one weight with a Day 1 dietary recall that was not completed. Participants with refused, "don't know," or missing responses to DSD010 are excluded because they cannot be assigned to the supplement-use or no-supplement-use group. Participants with missing or grouped sleep-duration values are excluded only from analyses of that numeric sleep outcome.

These counts describe the current analytic rules applied to the downloaded files. Any secondary analysis, such as an ingredient-specific melatonin analysis, will require its own eligibility counts and may have a smaller analytic sample.

Sources: NHANES DSQTOT_L documentation explains the use of WTDRD1 for 30-day supplement analyses and identifies zero weights with incomplete or uncompleted Day 1 dietary recalls. NHANES SLQ_L documentation documents the sleep-duration variables and their grouped values for less than 3 hours and 14 hours or more.